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ITAI 1371 Introduction to Machine Learning

13 Jun 2026

Reflection Journal

This lab helped me better understand the machine learning workflow and the different types of machine learning. The type that interests me the most is supervised learning it uses labeled data and provides clear results that can be measured. The wine classification project helped me see how models can learn from existing data and then make predictions on new data. It also showed me that different models can perform differently even when they are using the same dataset.

The most challenging part for me was understanding model evaluation and why one model performed better than the other. Before this lab I assumed that more complex models would always have better results. However, comparing logistic regression and decision tree models showed me that simpler models can sometimes perform better depending on the data. This helped me understand the importance of testing and evaluating models instead of making assumptions.

Since I work in Healthcare, I can see many ways machine learning could be used in my field. For example, supervised learning could help to predict patient risks or identify patients who may need additional medical attention based on their health records and test results. Learning about these concepts made me realize how machine learning can support better decision making in real world situations.

Overall, this lab helped me to connect the tools and concepts from previous lessons to the complete machine learning process. I would like to learn more about improving model accuracy, preventing overfitting, and how machine learning systems are used in healthcare organizations. I feel much more confident in my understanding of machine learning that I did before completing this lab.

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Reflection Journal: Module 3 Machine Learning Workflow

This lab helped me understand machine learning as more of a full process instead of just one step where a model gives an answer. Before this assignment, I mostly thought of machine learning as training a model and checking if it was right or wrong. After working through the Wine classification project, I can see that the model is only one part of the workflow. The data has to be loaded, explored, split, trained on, tested, and then evaluated carefully. That made machine learning feel more organized, but also more detailed than I first expected.

The main focus of this lab was using the Wine dataset to classify different wine classes based on numerical features like alcohol, malic acid, ash, color intensity, and proline. The dataset had 178 samples, 13 features, and three target classes. One thing that stood out to me early was that the dataset was clean, with no missing values. That made the lab easier to follow, but it also made me realize that real-world data probably will not always be this neat. In a real situation, checking for missing values and class balance would probably be a bigger part of the work.

One of the biggest things that clicked for me was the difference between supervised, unsupervised, and reinforcement learning. Supervised learning made the most sense in this lab because the dataset already had labels for each wine class. It was like the model had an answer key during training. Unsupervised learning is different because the model would be looking for patterns or groups without being told the correct answers ahead of time. Reinforcement learning also became clearer when I thought about it as trial and error, like a system learning from rewards and penalties. The examples in the assessment helped me understand that the type of machine learning depends on what kind of data is available and what the goal is.

The train/test split was another important part of the lab for me. At first, splitting the data seemed like just a setup step, but I understand now why it matters. The training set is what the model learns from, while the test set checks whether the model can handle data it has not already seen. That helped me understand the difference between memorizing and actually learning patterns. If a model only performs well on the data it trained on, that does not really prove it is useful. Testing it on unseen data gives a better

idea of whether it can generalize.

Comparing Logistic Regression and the Decision Tree was probably the most interesting part. Logistic Regression had the better accuracy at 88.9%, while the Decision Tree had 83.3%. Before this lab, I probably would have assumed the Decision Tree might do

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better because it seems more flexible and easier to visualize. But this showed me that the “better” model depends on the data and how well the model handles the problem.

A model that sounds more advanced or more flexible is not automatically better.

The confusion matrix also helped me slow down and think beyond accuracy. At first, I wanted to just look at the accuracy score and call that the result. But the confusion matrix showed where the model was actually making mistakes. Logistic Regression classified class_0 very well, but it had some confusion between class_1 and class_2. That made evaluation feel more realistic. Accuracy gives the overall score, but it does not explain the specific weaknesses of the model. I can see now why people use multiple evaluation tools instead of relying on one number.

The feature experiment was another good learning moment. I chose alcohol, color intensity, and proline as my three features. My model reached 83.3% accuracy, which was lower than the original Logistic Regression model. At first, that felt like I had picked the “wrong” features, but I think that was actually the point of the exercise. It showed me that feature selection matters a lot. Some features may seem useful, but the model’s performance depends on how well those features actually separate the classes. It also made me wonder how much better the model would perform if I tested more combinations or used all features with scaling.

One challenge I had was keeping all the steps connected in my head. The code itself was not impossible to follow, but there were a lot of moving parts: selecting features, splitting the data, training two models, calculating accuracy, reading the classification report, and then interpreting the confusion matrix. I had to remind myself that these steps are not random. They are part of a workflow. Each step answers a different question. What data do we have? What features are we using? How will we test the model? How well did it perform? Where did it make mistakes?

This lab also helped me connect machine learning to real-world applications. For example, in healthcare, supervised learning could be used to classify patients into risk categories based on lab results, symptoms, or medical history. That connection made the lab feel more useful because the same basic workflow could apply outside of the Wine dataset. The dataset changes, but the process is similar: collect labeled data, prepare it, train a model, test it, and evaluate the results carefully.

If I were improving this project, I would try using StandardScaler because some features have very different ranges. For example, proline has much larger values than features like alcohol or ash. Since Logistic Regression can be affected by feature scaling, standardizing the data might improve performance. I would also experiment with

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different feature combinations instead of choosing only one set. Another thing I would want to try is using more models, like k-nearest neighbors or random forest, to see how they compare.

Overall, this lab made the machine learning workflow feel much more real to me. I learned that building a model is not just pressing run and getting an accuracy score. It involves understanding the data, making choices about features, testing fairly, and interpreting the results. The biggest takeaway for me is that machine learning is not only about the final answer. It is about the process of figuring out whether the answer is reliable, where the model struggles, and what could be improved next.

Ved Dalwadi

In this journal entry, I will share my thoughts and experiences about the module 3 lab exercise. In this lab, I have built on the previous lab, working with machine learning models after learning the basics of ML models in Lab 3. In this lab, I learned about the different types of machine learning models, like Scikit-learn, Logistic Regression, and Decision Trees. This showed me how machine learning models are made and tested. The point of this reflection is to show what I learned in the assignment and how it can help me later on in the future.

The purpose of this lab is for me to learn about the different ways to teach a machine learning model. There were 3 types of machine learning that I learned about: supervised, unsupervised,

and reinforcement learning. The main type of machine learning in the lab was supervised, where the machine is trained on labeled data with correct answers, like studying with a practice exam and a key to check it. Unsupervised learning is something I believed to be the exact opposite of supervised learning, but now I know it just has a different goal in mind rather than supervised, as it doesn't have a correct answer and the machine has to find patterns without knowing if it's correct or not. I enjoyed reinforcement learning the most as the point system teaches the machine what to do and what not to do, like training a pet, where treats mean they did something good, and if not, they didn't they don't get a treat. These three types of machine learning helped me understand that, based on the data and problem, machine learning can differ very much.

Once I started the lab, I was shown three different types of machine learning. I was already familiar with reinforcement learning as I had seen it used to teach multiple skills to an AI, mostly including how to play a racing game and get the world record or a competitive time. However, it took me more time to understand supervised and unsupervised learning and their differences. I understand that both took data and gave an output. I didn't understand why one only used unlabeled data and the other used labeled data, and how that affected the outputs. As I worked through the lab, I saw the wine dataset being used for supervised learning, and the model was given the chemicals and the wine class, so when the test set came along, it was able to take the new information and translate that into one of its training sets to learn the patterns. Unsupervised learning takes unlabeled data and finds new patterns, whereas supervised learning is given a correct answer and can use that information to find patterns in new data. I first wondered how a computer is meant to give new information from this, but I realized that instead of basing it on previous information, it can see patterns to give correct or unseen information from before. This reminded me of module 2: in the iris dataset, the types of flowers were grouped, and I now realize that an

unsupervised model could best understand and break down the dataset to provide useful information.

This lab showed me the 6 steps in making artificial intelligence and how to take given data into predictions. However, while going through this process, 2 of the steps helped me realize that understanding how machine learning works is a very deep task that requires critical thinking. One of the steps that was interesting was the selection step, where I had to choose which 3 features the AI would learn from. I, however, believed that the model could get smarter with more information given, as it has more to work with. But when I tried to add more data to it, the model seemed to be less accurate. After thinking it over, I now understand that the more

information the model works with, the less accurate it may become due to the overload of information. Another step that surprised me is the model training, where both the Logistic Regression and Decision Tree ran, and I was surprised to see a 5% difference, as I believed that it was the same data, so the accuracy rating would be 1-2% different from each other. However, when the result was printed out, and I saw a 5% difference, this is due to the low depth of the decision tree and because Logistic regression works best with easily separated categories, which the wine dataset has. This made me realize that every part of the machine learning process needs to be carefully picked out; otherwise, the AI will not succeed.

When I was working through the lab, I had a few questions that I researched after the lab. One thing I thought about was what would happen if the decision tree had a depth of 13 instead of 3, if it could get a more accurate prediction. I then learned that if there was too much depth, the AI would just memorize instead of learning something new. Another thing I wanted to know was if all 13 features were used instead of just 3, as I believed that more data could make it more accurate, but as I was selecting the best data, I realized that the AI may get confused with all the different types of data.

Module 3 showed me the process of how the AI is made, trained, and learns, also how the person developing the AI has a lot of control over the AI, and if not done properly, the AI can not succeed in its development. This lab has taught me the complexity of making an AI and that it's a tedious task. In the future, I would like to understand how to increase the accuracy of a model, not just try until I get it right.